Inter-Datacenter Load Balancing for AI Inference

IETF 126

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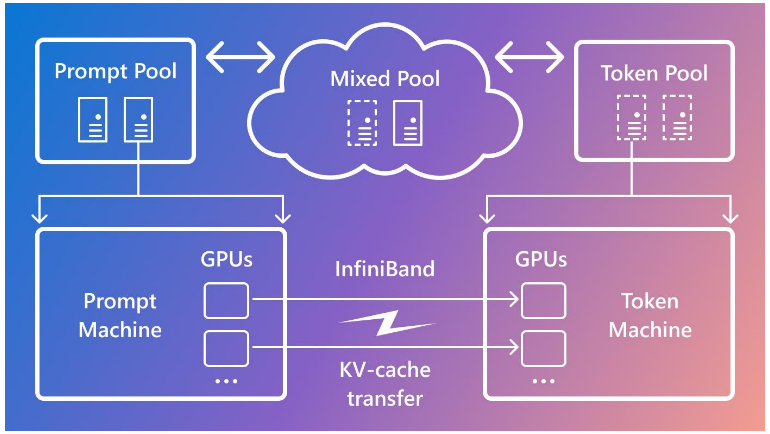
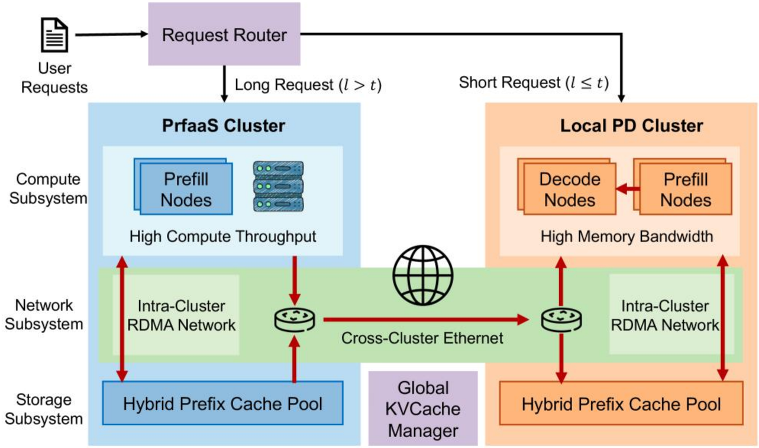
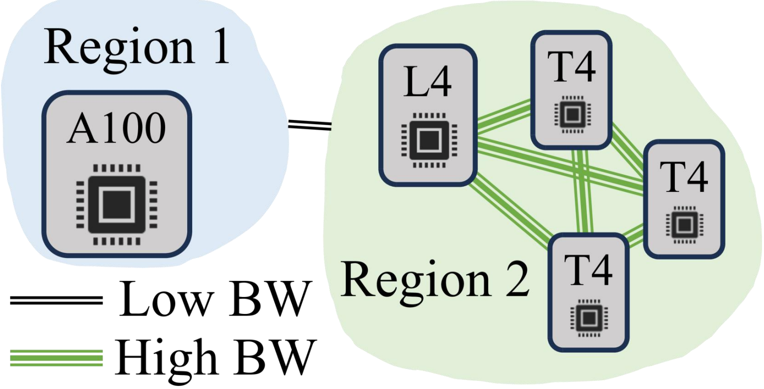
Why Inference Goes Cross-DC

Shift: Phase-specialized prefill and decode are increasingly deployed as separate pools.
Drivers: Long-context workloads widen the transfer-overlap window, while finite cluster capacity pushes inference beyond one RDMA island.







1 Phase asymmetry
Prefill is compute-bound; decode is memory-bandwidth-bound. Separate pools use the right hardware for each phase.

2 Long-context opportunity
Longer prefills create a larger compute window; layer-wise KV generation can overlap with cross-cluster transmission.

3 Capacity pressure
Demand and hardware heterogeneity exceed one cluster's pool. Cross-DC deployment exposes remote compute and elasticity.

Sources: Splitwise (Microsoft Research, 2024); PrFaaS (2026); Helix (ASPLOS 2025).

Why Cross-DC Inference Is Worth It

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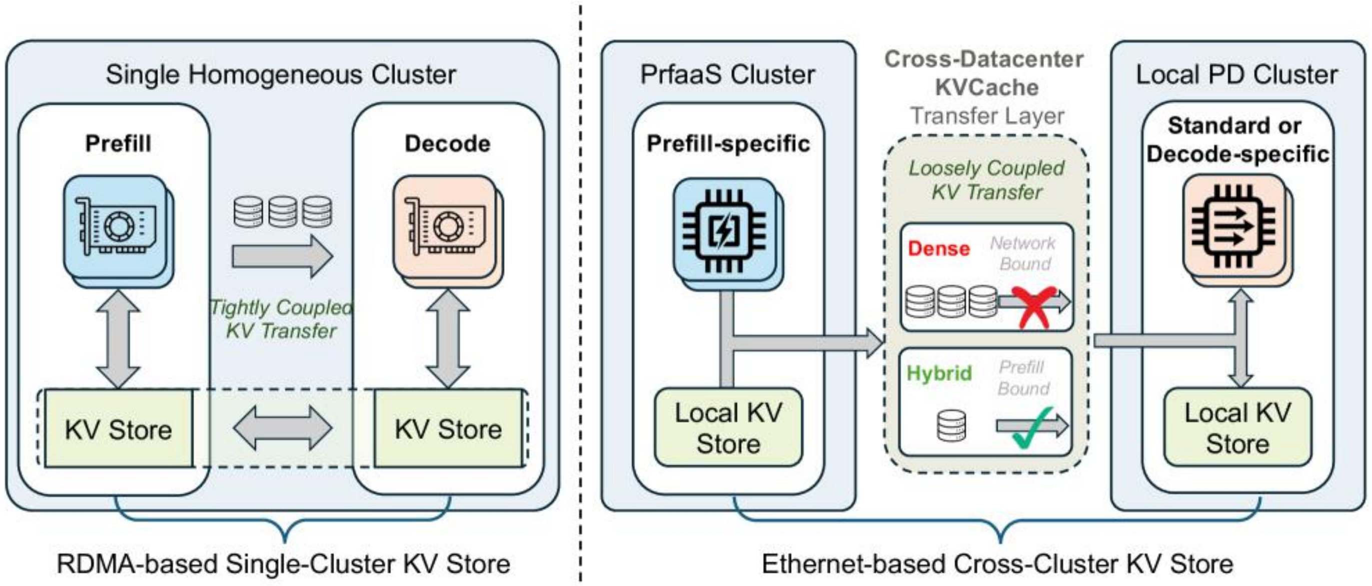
Architecture: Remote compute-dense prefill -> pipelined KVCache transfer -> local memory-optimized decode.
Boundary: Selective offload and network-aware scheduling must keep transfer cost below the prefill-side gain.

Cross-datacenter PD disaggregation



1 Resource pooling
Use non-colocated accelerators across clusters and regions.

2 Phase-specific hardware
Compute-dense prefill; memory-bandwidth-optimized decode.

3 Independent scaling
Tune the P:D capacity ratio as traffic and cache-hit rates change.

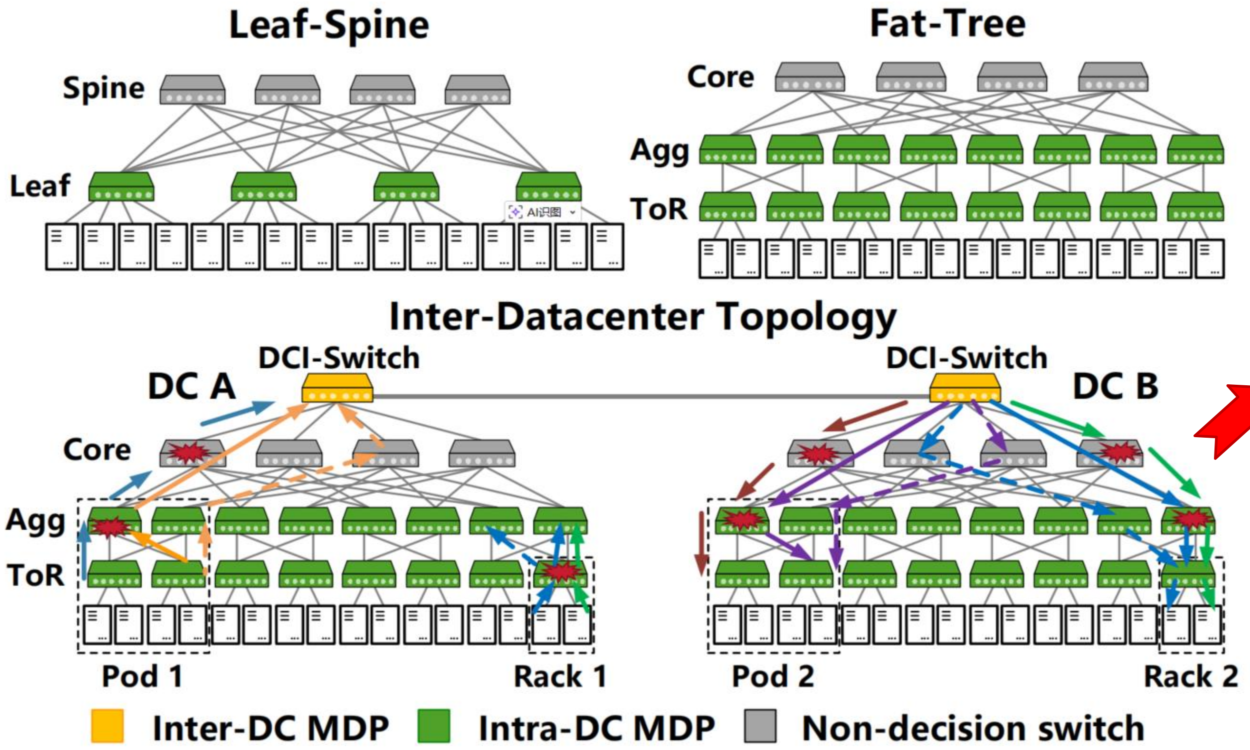
4 Better long-context SLOs
Accelerate remote prefill while keeping token generation near users.

5 Measured case study
+54% throughput, -64% P90 TTFT; about +15% throughput at equal cost.

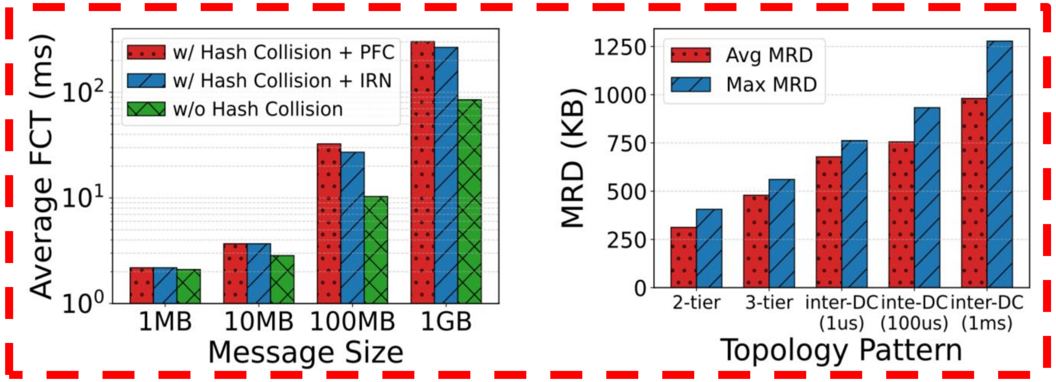
Source: Qin et al., Prefill-as-a-Service, Fig. 1 and case-study results, arXiv:2604.15039v2 (2026).

Cross-DC KV Transfer Challenges

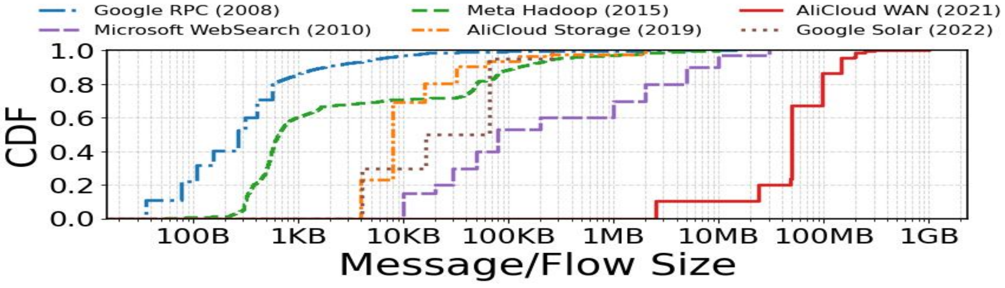
False congestion: LB mistakes create apparent congestion despite available capacity.
Path constraints: DCI switches introduce new load-balancing decision points.
Traffic distribution: Cross-DC traffic carries larger messages, with KVCache transfers reaching hundreds of MBs or more.



Inter-DC Network Topology



Path conflicts and packet reordering increase



Inter-DC Message Sizes

Why Existing Solutions Fall Short

Flow-level: ECMP is congestion-oblivious, while heterogeneous RTTs can skew SGLB's congestion estimates and path selection.
Subflow-level: RDMA cannot reliably detect flowlets and needs packet-reordering support.
Packet-level: Long propagation delays make packet reordering increasingly expensive.

Scheme / Metric		Congestion Awareness	Reordering	LB Efficiency	Cross-DC LB
Flow-level	**ECMP**	×	None	Low	×
	SGLB (SIGCOMM '26)	✓	None / Low	High	×
Subflow-level	**E-ECMP (SIGCOMM '24)**	✓	None	Medium	×
	UnoLB (SC '25)	✓	Medium	Medium	✓
Packet-level	**AR (NVIDIA '23)**	✓	High	High	×
	REPS (EuroSys '26)	✓	High	High	×
Hybrid	**Strata**	✓	**Low**	**High**	✓

Design: Hierarchical Load Balancing

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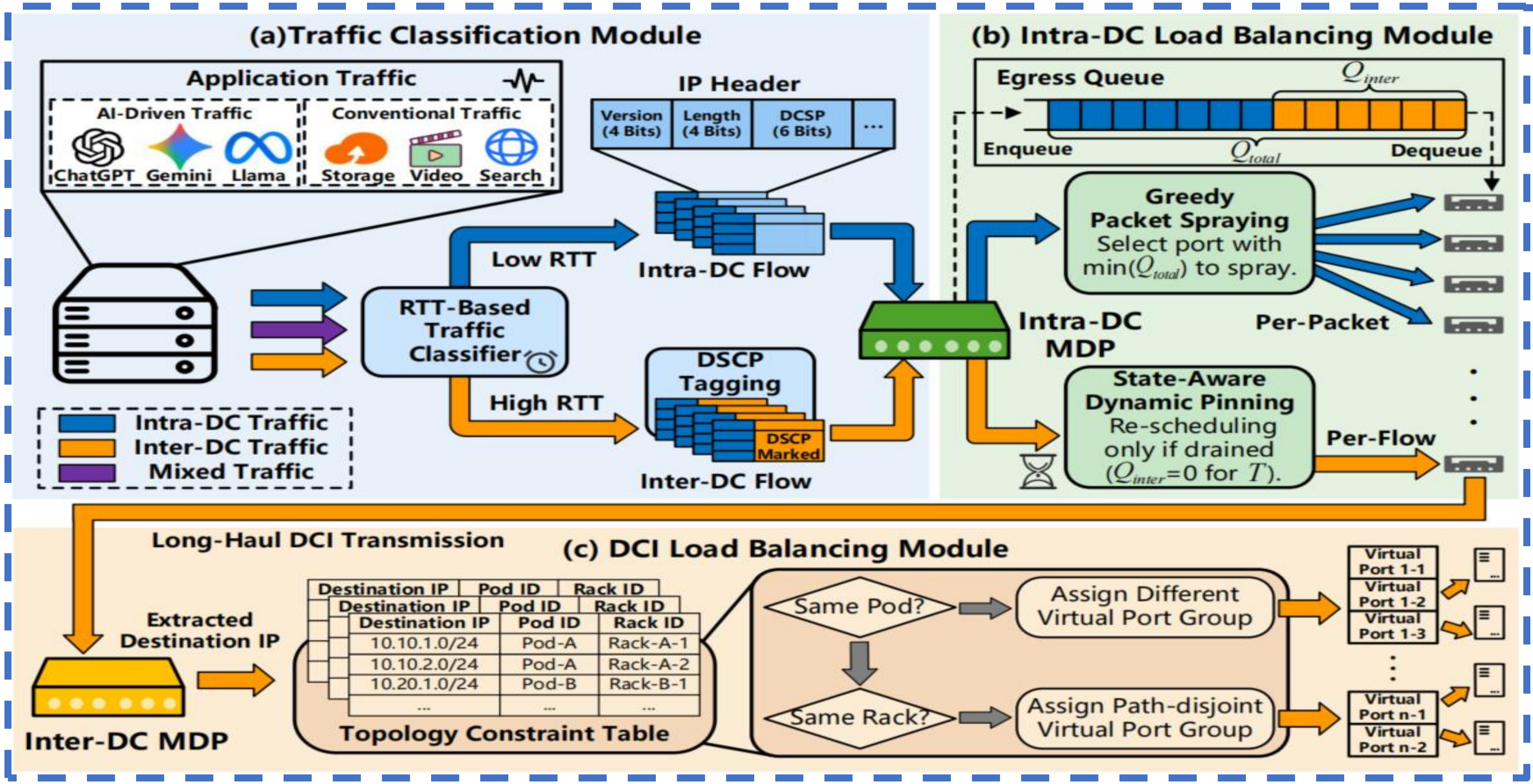
Core idea: Use different routing policies for intra- and inter-DC traffic.
Intra-DC: Shortest-queue packet spraying at packet granularity.
Inter-DC: Dynamic per-flow path binding with downstream path-disjoint constraints.



Traffic Classification
- ✓ RTT-based flow detection
- ✓ Separates intra-/inter-DC traffic

Intra-DC Balancing
- ✓ Shortest-queue packet spraying
- ✓ Per-packet forwarding

Inter-DC Balancing
- ✓ Dynamic per-flow path binding
- ✓ Downstream path constraints

Evaluation: Large-Scale Simulation

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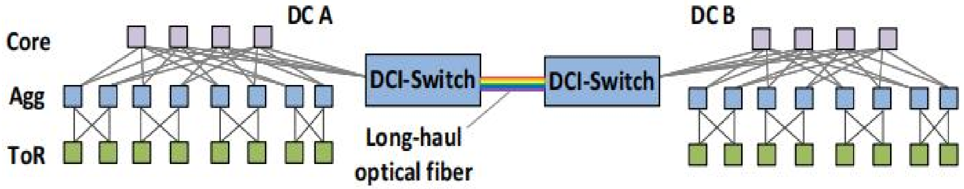
Implementation:

- ✓ NS-3 simulation;
- ✓ DCQCN as the default congestion control

Topology:

- ✓ 800-Gbps intra-DC links; four servers per ToR
- ✓ 1:1 oversubscription; 1-ms DCI delay (~200 km)

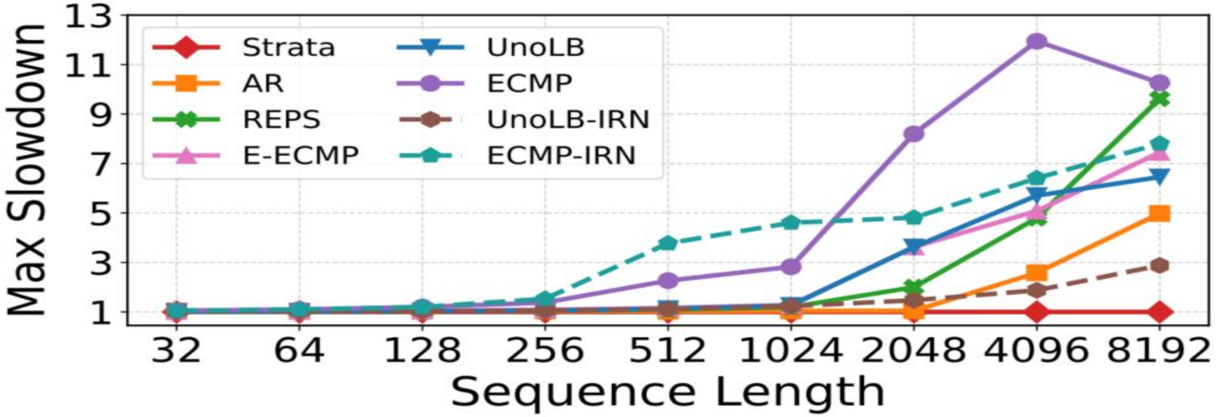


Traffic:

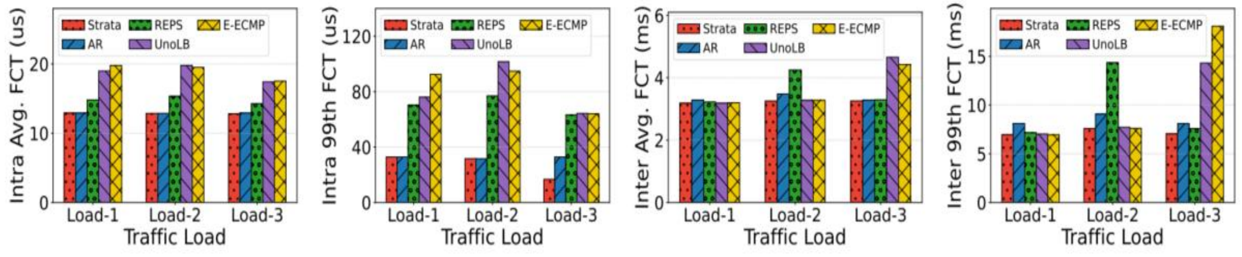
- ✓ Pipeline-parallel (PP) traffic
- ✓ Mixed RPC/storage: Google RPC + Alibaba WAN (Load-3)
- ✓ Mixed storage: Alibaba storage + Alibaba WAN (Load-1: 0.3 + 0.4; Load-2: 0.2 + 0.5)

PP Traffic

Mitigates path conflicts



Mixed Traffic



(a) Intra-DC average FCT (b) Intra-DC 99th FCT (c) Inter-DC average FCT (d) Inter-DC 99th FCT

Tail FCT drops 73.9% intra-DC and 61.2% inter-DC

Strata keeps the best intra-DC mean FCT while reducing tail latency for both intra- and inter-DC transfers

Future Work

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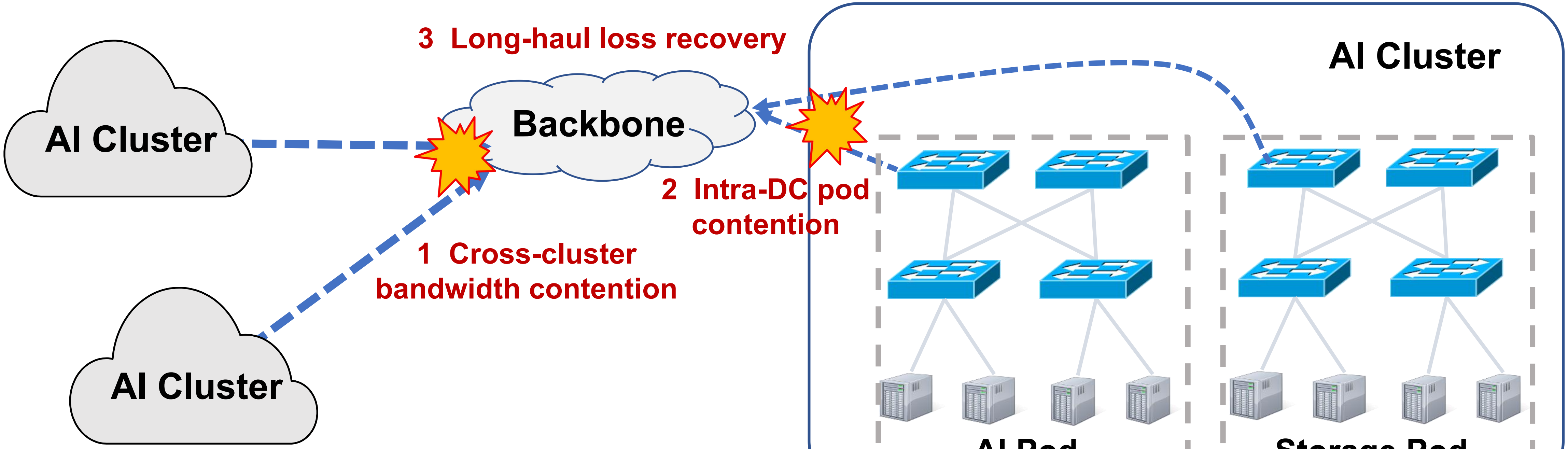
Multi-Cluster AI Networking:

- ✓ Multi-directional AI traffic competes for backbone bandwidth.
- ✓ Pods within the same AI DC compete for fabric resources, requiring bandwidth guarantees for AI traffic.
- ✓ Random loss inflates tail latency under selective retransmission.



Multi-Cluster AI Networking Challenges

Thank You!

Discussion and Feedback Welcome

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